



HYPERLOOP | IN REVIEW

July 2025

17 JULY 2025

HYPERLOOP'S EVOLVING LANDSCAPE

2025-H1

hyperloop
development
program

Preface

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Senator Anjes Tjarks of Hamburg (left) and Domenik Radeck of TUM (right) at UITP Summit

Dear **Hyperloop** enthusiast,

The year 2024 laid the basis for 2025 being an absolute key year for Hyperloop. The recognition of **Hyperloop** in the **TEN-T Regulation** made the start, key European political leaders such as **Ursula von der Leyen** and prominent figures like **Mario Draghi** have projected **Hyperloop** to the forefront of the political agenda: The Commission President has charged her new Commissioner for Transport, **Apostolos Tzitzikostas**, to promote innovative technologies such as **Hyperloop** and Mario Draghi in his report on European competitiveness has pointed to **Hyperloop** as being one of the innovative technologies that could bring Europe's economy forward.

At the beginning of 2025, the new German government included a commitment to **Hyperloop** in its coalition treaty and shortly after, the newly elected government of the City of Hamburg did the same. These events suddenly bring Germany to the forefront of **Hyperloop** development.

At the same time, the Italian project in the **Veneto** region has finalized its feasibility study and all actors are now discussing the next steps. There is also progress in Spain on the project in the **Castilla La Mancha** region. The test facility go-Tube in **Lower Saxony** in Germany has been inaugurated and started to work. **Nevomo** has secured new projects. Companies who are not part of **HDP** such as **Swisspod** and **Transpod** also report progress.

The **Hyper4Rail project** and the very large consortium implementing it is on a good way and reports first results. A further study initially proposed by the European Parliament is about to be launched by the European

Commission and will allow **Hyperloop** promoters to work on demonstrating the safety concepts developed by them.

It can be expected that the increased political support will lead to further financial resources being mobilized by the public and private sectors and allow **Hyperloop** development to advance more quickly.

Therefore, it can be expected that this year will be a very good one for **Hyperloop** and bring the technology and the demonstration of its business case forward.

At the same time, the **HDP** is undergoing significant changes. The aim is to empower and transform it to become a comprehensive voice and an even stronger advocate for **Hyperloop**.

This is achieved in two ways: attracting new members to make it representative for the whole **Hyperloop** ecosystem and develop its lobbying role with political decision makers. In order to be able to attract more members, its governance structure has been adapted to make it more transparent and to offer more added-value to members. It has been possible to onboard several new members and promising talks are underway with more potential partners.

A lobbying strategy has been developed and is implemented that aims at reaching more decision makers at European and at national level. Many of them have already been contacted and more actions are under way. In addition, two important events have been organized in the first half of the year, one in Brussels and one in Hamburg and more are planned.

With the changes within **HDP** as outlined above, **HDP** becomes a more credible interlocutor for authorities and will be able to achieve more impact. It is on a good way to become a comprehensive branch organization.

All this only became possible thanks to the very good collaboration with our partners and their relentless support for the activities proposed! Therefore, I wish to thank all our present and future partners for their trust and support! Together we are showing a positive and convincing picture of **Hyperloop** based on the Vision Paper we agreed in 2024.

Sincerely,

Klaus Rudischhauser





1. TOWARDS A NEW TRANSPORT TECHNOLOGY

Hyperloop is being developed as a **high-speed guided transport** and can significantly **improve connections across regions and countries**. Notably, being **energy efficient** during operations, the new transport mode will contribute to **reducing energy consumption** in transport and mobility on long distances.

Utilizing existing technologies

The hyperloop combines existing technologies of magnetic levitation trains and a low-pressure environment in tubes. A hyperloop vehicle can therefore move without rolling resistance and with very limited air resistance. The tubes have a low-pressure environment maintained by air pumps. This enables movement with minimal aerodynamic resistance and low energy consumption. The system can be scaled up for high-frequency transport, which is comparable to the efficiency of today's metro or light rail connections.

Enabling hybrid transportation

Hyperloop is designed for passengers and cargo. Vehicles, resembling small airplanes without wings, move through a tube. The vehicles use magnetic levitation, guidance, and propulsion, which consists of steel rails in the tube and magnets on the vehicles. Safe and efficient access for people and goods to the vehicle is being designed through applying air docks.

Enhancing multimodal solutions

Hyperloop transport requires seamless integration within a multimodal network to reach its full potential. Transport hubs will facilitate the boarding of passengers or the transfer of cargo, as well as the transit to intercontinental aviation at airports and first- and last mile solutions within distribution centers.

Strengthening continental & regional transport

Within a European network the cruising speed is expected to be 700 km/h, according to the HDP's Hyperconnected Europe report (2022). Therefore, the hyperloop has the potential to alleviate transport over long distance, such as the strain on busy airports and congested motorways or railway networks. For different sorts of distances, the speed can be easily adjusted, making the hyperloop a versatile transportation solution for connections across a continent or in densely populated metropolitan regions.

Lowering costs with hyperloop

The hyperloop infrastructure's construction costs are comparable to high-speed rail due to its modular structure and relatively small spatial footprint, based on a CAPEX comparison executed by Deutsche Bahn E.C.O. Group and Hardt Hyperloop in 2022. In addition, the hyperloop's efficiency is increased by the friction-less magnetic propulsion and levitation as well as an enclosed environment, protecting the moving vehicles from environmental conditions. These features result in less wear and tear and, as a result, lower maintenance costs.

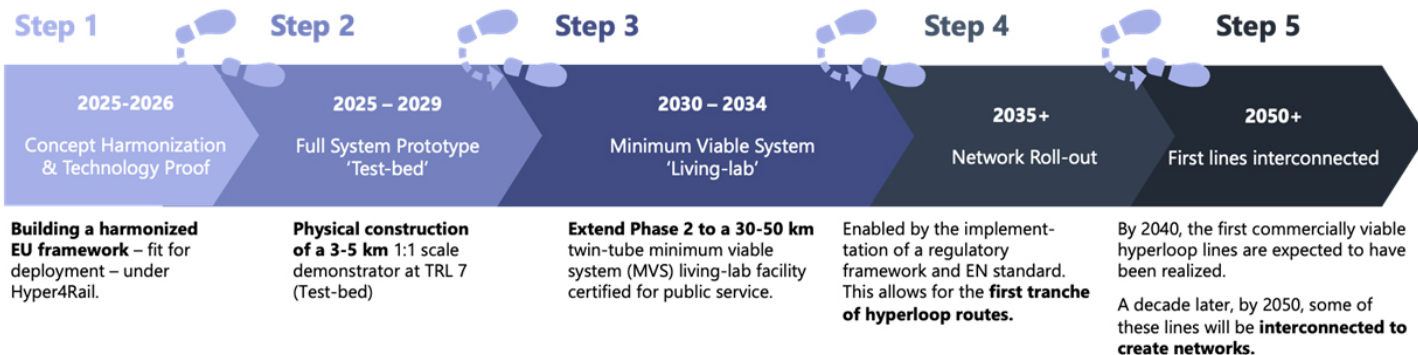
INDUSTRY ALIGNMENT THROUGH HDP VISION PAPER - A COORDINATED PATHWAY

After a decade of global R&D, Europe now accelerates towards implementation.

From 2025–2026, existing test facilities (NL, CH, DE, ES, TR) will align technology via the **Hyper4Rail project**. By 2029, a full-scale prototype (**TRL7**) will prove system feasibility. Between 2030–2034, a 30–50 km Minimum Viable

System will demonstrate real-world operations up to 1000 km/h. By 2035, commercial rollout begins—building on the MVS foundation, backed by EU standards and regulatory readiness.

Further reading in the HDP Vision Paper ([Link](#))



**goTube test facility
in Emden, Germany**



2. HYPERLOOP DEVELOPMENT IN H1- 2025

A chronological overview of events from **January** until **July 2025**:

JANUARY

- **Lobbying**: a comprehensive lobbying strategy including a mapping of all relevant services in Brussels was drafted and implementation started. Letters were written to all parties in Germany with a view to the upcoming federal elections, drawing the attention of **Members of Parliament to Hyperloop**.

- **HDP**: a new membership structure was implemented. Partners can now be core partners, participate in the steering committee and the board. Or they can be associate partners without participation in the governance. A membership fee was introduced that would allow **HDP** to have some independent own income.

- A spin-off of the research program at the **Technical University of Munich** was founded to push the commercialization of the technology forward. The new company is called **Neoways**.

- **Deutsche Rail Operations** and **DLR** started the 18-month study on assessment of the current state of development of **Hyperloop concept**.

FEBRUARY

- **Lobbying**: a comprehensive document on expectations of the **Hyperloop** community regarding the new Commission was developed and presented at the Commission workshop on 13 of February. **HDP** sent letters to the Commission President and several Commissioners outlining the significant impact that **Hyperloop** could have on European industrial competitiveness.

- **DG Move** held a workshop with Hyperloop promoters to finalize the **Ramboll** study under way and discuss its future activities on **Hyperloop**.

- Planning started with **HDP** partners for a joint stand at the UITP Summit in Hamburg 16-18 June.

- Construction of a large test tube by **Swisspod** in Colorado underway.

- The **Global Hyperloop Competition 2025** was hosted by the **Indian Institute of Technology, Madras**, introducing the Hyperloop to India, providing a global platform for innovative teams.

MARCH

- **Inauguration of the Go-Tube test facility** at the **University of Emden/Leer**.

- The coalition treaty of the new **German Federal Government** contains a commitment to **build a Hyperloop reference line**.

APRIL

- The coalition treaty of the new City Government of Hamburg says: **Hamburg is interested in the Hyperloop technology and will together with its partners propose to the Federal Government to build a first technical reference line**.

- **HDP** is approached to lend support to discussion in the United Kingdom to promote **Hyperloop** with the UK government.





2. HYPERLOOP DEVELOPMENT IN H1- 2025 (CONT.)

MAY

- **HDP** together with **TUM** organized a public event at the **Representation of the Free State of Bavaria** in Brussels on “**Hyperloop, Sustainable Transport for the future**” with **MEP Markus Ferber** and the **Head of Cabinet of Commissioner Tzitzikostas**. Around 200 participants attended. TUM prototype vehicle showed in the patio of the Bavaria representation.
- The **Korean government** launched a **task force on April 9** to develop the country's own **Hyperloop technologies**, investing **12.7 billion won** (\$8.8 million) for initial research.

JUNE

- **HDP** present at the **UITP Summit in Hamburg** with a joint stand attended by **TUM** with its prototype vehicle and around **10 partners of HDP**. Public presentation at the stand of **Max Bögl**, new member of **HDP**.
- A **special session** is held with the **Senator for Mobility of the City of Hamburg** and a collaboration is agreed, by which **HDP** will accompany and support the project.
- **HDP** sent a letter to **DG Move** arguing against the hesitant position taken by **DG Move**. The letter is co-signed by a number of **HDP** partners to give it more traction.
- The feasibility study for a **10 km test track for the Hyper Transfer ultra-fast transport system** in **Veneto region, Italy**, has been successfully concluded.





2. HYPERLOOP DEVELOPMENT IN H1- 2025 (CONT.)



Delegation at Brussels with Mr. Markus Ferber (MEP) and Ms Anna Panagopoulou, head of Cabinet Sustainable Transport and Tourism

Hyper4Rail project consortium at the kick-off



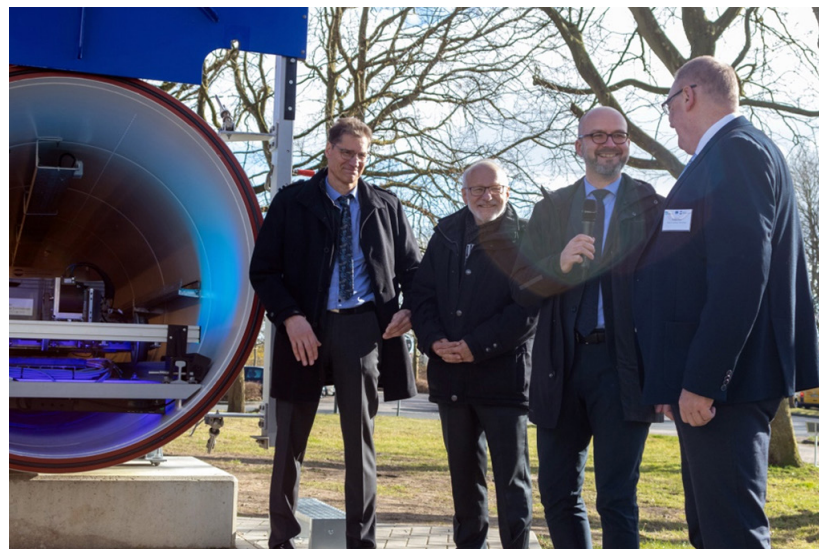


2. HYPERLOOP DEVELOPMENT IN H1- 2025 (CONT.)



Inauguration of goTube at Hochschule Emden/Leer, Institute of Hyperloop Technology, with Mr. Eichel (Ministry of Science and Culture of Lower State of Saxony), and Prof. Dr. Kreutz (President of Hochschule Emden/Leer)

Delegation at UITP in Hamburg with Mr. Anjes Tjark, Senator for Transport and Mobility Transition Hamburg



2. HYPERLOOP DEVELOPMENT IN H1- 2025 (CONT.)



Global Hyperloop Competition in Chennai (February), with Zeleros, Hardt, Neoways, Erciyas and TuTr attending

Visit of Tongji University and the Southwest Jiaotong University from China to the European Hyperloop Center



A large white cylindrical test facility is being lifted by a crane. The crane is white with green accents and has the website 'www.consensa.ch' printed on its boom. The facility is being hoisted by a red lifting strap. In the background, there is a white rectangular structure with a circular opening. A worker in a yellow safety vest and white hard hat is standing near the facility. The scene is set outdoors on a paved area with a blue sky and clouds in the background.

Construction of DemoTube test facility in Dübendorf, Switzerland



3. HYPER4RAIL STARTED – DELIVERING A EUROPEAN HARMONIZED CONCEPT

The Hyper4Rail Consortium, coordinated by HDP, unites 27 public and private organizations advancing Hyperloop innovation. With diverse participants from the private and public sector, the consortium shares a unified goal: creating high-speed, sustainable transportation that aligns with the EU's vision of a seamless and integrated European network.

As the Hyperloop technology is still in its early stages and there is a lack of standardisation across developers, Hyper4Rail offers the opportunity to build consensus on industrialisation, safety, regulatory, and technical frameworks and convergence on design considerations through a structured, all-stakeholder approach.

The project is focused on Hyperloop design definitions at Technology Readiness Level (TRL) 2, which involves developing and confirming the basic principles of Hyperloop systems such as propulsion and vacuum technology. The process is followed by validation of key subsystems at TRL4, the level of testing and feasibility demonstration in laboratory environments. A safety framework, operational guidelines, and business cases will be developed to align Hyperloop with Europe's multi-modal transportation goals under the Trans-European Transport Network (TEN-T). Ultimately, the project will produce a unified roadmap to guide the deployment of Hyperloop systems across Europe.

The project uses an iterative design process, allowing for continuous refinement as ideas and designs evolve. Social science disciplines will also be incorporated to assess societal acceptance, economic impacts, and environmental sustainability. Hyper4Rail is structured into six work packages, focusing on design, validation, business modelling, and the development of a detailed roadmap for implementation.

Once the Hyper4Rail project is completed, attention will shift toward large-scale demonstrators, further collaboration within the industry, and pilot implementations. The roadmap developed during the project will help guide future research and development investments, as well as inform policy adjustments.

**THE GOAL IS FOR HYPERLOOP TECHNOLOGY TO BE INDUSTRIALISED AND DEPLOYED
AT SCALE BY THE 2030S**



3. HYPER4RAIL STARTED – DELIVERING A EUROPEAN HARMONIZED CONCEPT (CONT.)

1

Deliver a harmonized Hyperloop concept

2

Identify use-cases for Hyperloop and investigate its business case

3

Validate key Hyperloop Sub-systems

4

Provide a Roadmap for further industrialization and implementation



Grant Notice

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4. HYPERLOOP TEST FACILITIES IN 2025

OPERATIONAL FACILITIES

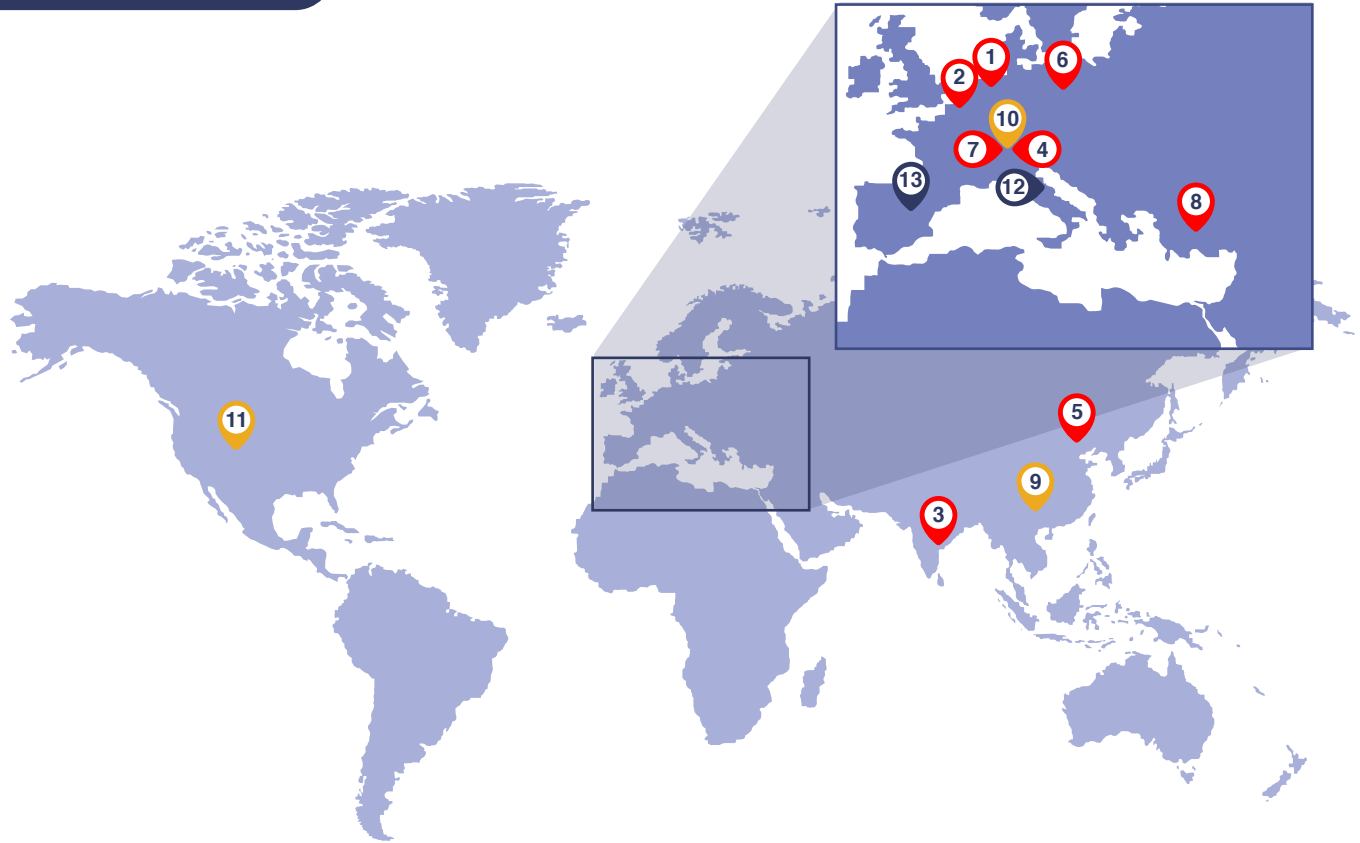
1. European Hyperloop Center (NL)
2. goTube test facility (DE)
3. ITT Madras test track (IN)
4. TUM Hyperloop test track (DE)
5. Datong Shanxi test track (CN)
6. Nevomo test track (PL)
7. Swisspod test track (CH)
8. Tubitak (TR)

UNDER CONSTRUCTION

9. Chengdu test track (CN)
10. DemoTube (CH)
11. Swisspod test track (USA)

PLANNED

12. Veneto (IT)
13. Ciudad Real (SP)



A photograph of a hyperloop test track. The track is a long, cylindrical tunnel with a concrete interior. The walls are lined with multiple rows of bright, blue LED lights that create a strong perspective effect, drawing the eye towards the center of the tunnel. In the middle of the track, a sleek, aerodynamic vehicle is positioned. The vehicle has a white nose cone and a dark, possibly black, body. It is resting on a set of metal tracks that run along the bottom of the tunnel. The overall atmosphere is futuristic and high-tech.

**Hyperloop test track
and vehicle at Technical
University Munich (GER)**



5. HYPERLOOP IN POLICY MAKING

The **first half of 2025** is marked by several **important advances at the political level** which promise that **Hyperloop** will receive a **higher level of attention for the rest of the year**. Most of these have happened in **Germany**.

DEVELOPMENTS AT EU LEVEL

- The Commission has abandoned the impact assessment destined to make the case for a regulatory framework given that the consultant could not deliver the assessment in the necessary form. The study will be published soon and the Commission will have to make a decision on how to proceed. At the same time, the Commission is preparing to tender the PPPA Project granted by the European Parliament. The Directorate for Transport (DG Move) is waiting for the outcome of these studies and those of Hyper4Rail, before taking any further action on Hyperloop. HDP is actively trying to convince DG Move to take a more proactive role.
- The new Transport Commissioner has not yet put forward his ideas for action on Hyperloop.
- HDP is actively trying to get Hyperloop into the work program for increasing industrial competitiveness announced for redressing the European economy. This includes also positioning Hyperloop in the research and development programs under the next multi-annual financial framework.

COUNTRY SPECIFIC DEVELOPMENTS

GERMANY

- Following the federal elections in February, a government of the conservative and the social-democrat parties is formed. In the coalition treaty a mention to constructing a Hyperloop reference line is made. In addition, magnetic levitation technologies have been granted access to specific funding sources destined for local authorities.
- In Hamburg a new city government composed of the social-democrat and the green party is formed. In the coalition treaty the building of a Hyperloop reference line is mentioned. HDP had the opportunity to meet the Senator for Mobility and to agree a collaboration. Talks are in progress with the state of Schleswig-Holstein to build a commercial line to Kiel.
- In Bavaria, the state government continues to actively support the construction of a hyperloop reference track. With funding from the Bavarian parliament, the Technical University of Munich is now in the advanced stages of a feasibility study, selecting and evaluating potential locations and preparing for the planning and implementation phase.
- The Eisenbahnbundesamt has commissioned a study to establish the state of the art in Hyperloop technology.



5. HYPERLOOP IN POLICY MAKING (CONT.)

COUNTRY SPECIFIC DEVELOPMENTS

SPAIN

• Preparations for the tendering of a project in Castilla La Mancha are going forward. The fact that this process has been moved to the level of the central government is a sign for a certain political interest.

ITALY

• Currently the results of the feasibility study carried out by the Consortium of the Hyper Transfer Project are evaluated and the results will determinate the future of the project.

INDIA

• The government is pushing forward its ambitious plans for Hyperloop, best demonstrated by the attendance of the Railway Minister at a recent conference on Hyperloop.

USA

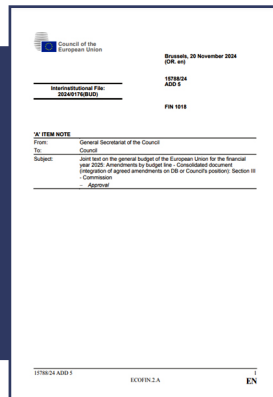
• A significant test track is currently finalized in Colorado with a certain amount of support by the government.

CANADA

• A project linking various capitals of provinces is being pushed forward.

Hyperloop currently is the only potential transport mode under development allowing to transport passengers and goods with low energy usage and without any pollution (i.e. GHG emissions and noise).

In order for the **trans-European transport network to keep up** with innovative technological developments and deployments, **sustainable emerging technologies that enhance and ease the transport and mobility of passengers and freight should be encouraged.**



**European Hyperloop
Center test facility in
Veendam, the Netherlands**





6. FUTURE OUTLOOK

In **2025, major breakthroughs** are possible based on the **changing political environment**, in particular in **Germany**:



• The City of Hamburg has submitted a significant funding request for federal funds to develop a vehicle and is seeking to secure funding for a test track of several kilometers. A consortium is currently being established and HDP has secured a supporting role in the project.



• The Hyper4Rail project, funded by the European Union and Europe's Rail, is expected to publish its first results in 2025-H2.



• The European Hyperloop Week has taken place in July, during which university teams present prototype components and pods to a panel of judges, competing in several categories. This edition is organized at the European Hyperloop Center in Veendam, the Netherlands.



• We are hoping to see the presentation of the EU-led action plan for hyperloop. We are keen on learning what the European Commission will propose for preparing next steps in the European legislative process.



• Following the initiative of the European Parliament in 2024, we expect that public funding will be made available for hyperloop research aimed at safety of passengers, which is an opportunity for the EU to maintain its leading role in standardization of new technologies globally.



• TWe hope to secure significant R&D funds from FP10 under the new multi-annual financial framework of the EU.



• TWe look forward to the follow-up of the first part of the Hyper Transfer study in Italy, aimed at connecting the cities of Venice and Padua with hyperloop.



• We expect the DemoTube test facility in Dübendorf, Switzerland - of which the first section has been realized - to be expanded to a 120-meter-long tube.



• HDP will continue to evolve into an efficient and effective branch representation with broader membership, a renewed governance structure and further financial support from the Dutch government.



SUPPORTED BY



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Contact: general@hyperloopdevelopmentprogram.com



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